

Healthcare Appointment Management System

Modular, scalable, and bilingual Healthcare Appointment Management System is organized into three verticals (Patient Mobile App, Patient Web App, and Healthcare Facility App).

Healthcare Facility covers:

- Clinic
- Polyclinic
- Pathology Lab
- Radiology Center
- Hospital
- Physio Clinic
- Skincare & Laser Center

System Structure:

1. Patient App (Mobile)
 - Built in Flutter for Android & iOS.
 - Designed for easy patient onboarding and booking on-the-go.
2. Patient App (Web)
 - Built in NextJs for browser-based access.
 - Same features as the mobile app but optimized for desktops/laptops.
 - Highly optimized for search engines & GPT
3. Healthcare Facility (for Doctor, Technician, & Admin Dashboards)
 - Built in React.js.
 - Includes doctor-facing, technician facing and facility-facing dashboards for managing appointments, earnings, and patients.

Features:

Feature	Patient App (Mobile)	Patient App (Web)	Healthcare Facility App
Core Patient Management			
Patient profile creation	✓	✓	✓
Patient info management	✓	✓	✓
Family account management	✓	✓	✗
Patient notes & history access	✓	✓	✓

Doctor, Technician & Facility Management			
Doctor profile (availability, fees)	✓ (view)	✓ (view)	✓ (manage)
Facility profile	✓ (view)	✓ (view)	✓ (manage)
Facility ratings & reviews	✓ (add/view)	✓ (add/view)	✓ (view)
Multi-doctor team calendar	X	X	✓
Appointments & Scheduling			
Appointment booking	✓	✓	✓
Appointment rescheduling/cancellation	✓	✓	✓
One-tap rebooking (follow-ups)	✓	X	X
Queue management	X	X	✓
Web check-in	✓	✓	✓ (View Status)
Calendar sync (Google, iCal, Outlook)	✓	✓	✓
AI-based scheduling assistant	✓	✓	✓
Overbooking control (Race condition, buffer time, emergencies, walk-ins)	✓	✓	✓
Multi-location booking for facilities with multiple branches	✓	✓	✓
Payments & Billing			
Digital payment integration	✓	✓	X
Earnings dashboard	X	X	✓
Analytics dashboard (most booked times, patient demographics)	X	X	✓

AI Features			
AI doctor suggestion	✓	✓	✗
In-app symptom checker	✓	✓	✗
Communication & Notifications			
Automated SMS/email/WhatsApp reminders	✓	✓	✓
Notifications & push alerts	✓	✓	✓
System & Language Support			
Bi-lingual UI (English & Arabic RTL)	✓	✓	✓

Additional Cost (Estimated) :

- Cloud Hosting (Cloudacropolis / ODP): ~ OMR 60–120 / month
- SMS & WhatsApp Gateway Fees: ~ OMR 0.010–0.030 per SMS / message
- Payment Gateway Charges (Thawani / Ompay): Typically 2–3% per transaction
- Email Services (SendGrid / AWS SES): ~ OMR 10–20 / month depending on volume
- App Store (iOS/Android) Developer Accounts: One-time ~ USD 100 (Google) + USD 99/year (Apple)

Deliverables:

1. Mobile Patient App – Android & iOS.
2. Web Patient App (React.js) – browser-based patient access.
3. Healthcare Facility App (React.js) – Doctor & Admin dashboards.
4. Backend (NestJS/Express) with REST APIs.
5. Database (MongoDB) with proper indexing & backups.
6. Integrations: Calendar, SMS/Email/WhatsApp, Payment Gateway.
7. Deployment Setup on Cloudacropolis / ODP.
8. Documentation: API docs, admin usage guide.

Next Steps:

- **Kick-off Meeting:** Align scope, milestones, and communication protocol.
- **Detailed user journey documents** for each role (patient, doctor, technician, and facility).
- **Agile sprint plan** with timelines, deliverables, and acceptance milestones.
- **Technical architecture & execution plan** with integrations mapped out.
- **UI/UX designs or wireframes**
- **Design Phase:** Finalize logo, colors, fonts and design artifacts.
- **Development (Agile Sprints):** 8-week sprint cycle with weekly demos.
- **UAT & Feedback:** Client reviews and approve deliverables.
- **Deployment:** Deploy Cloudacropolis/ODP and publish mobile apps.